FYP
Proposal

Emotion-Aware Emergency Dispatch System

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Summary

- Problem Statement
- Business Objectives
- Project Scope
- Project methodology
- Project Schedule (Time lines)
- Budget / Costing
- Project tools or H/W required
- FYP Deliverables
- References

1- Problem Statement

- Huge Staff Shortages: Most 911 centers are missing over 80% of the people they need. The few workers left are totally exhausted.
- Too Much Pressure: Because they're overwhelmed by calls, it's easy for operators to make mistakes or take longer to send help.
- Missing the Big Picture: Since they can only hear audio, it's hard for dispatchers to judge how bad a situation really is based on a caller's tone.
- Too Much Manual Work: Typing everything in by hand wastes precious seconds that people in an emergency don't have.

2- Business Objectives

- Operational Efficiency: Handle 30% more calls with existing staffing levels by automating data entry and triage.
- Cost Reduction: Eliminate recurring Cloud API fees by pivoting to a one-time Local GPU investment.
- Performance Targets:
- Latency: Transcription < 2 seconds.
- Uptime: 99.9% (Fully offline-capable).
- Accuracy: >80% alignment with human emotion consensus.

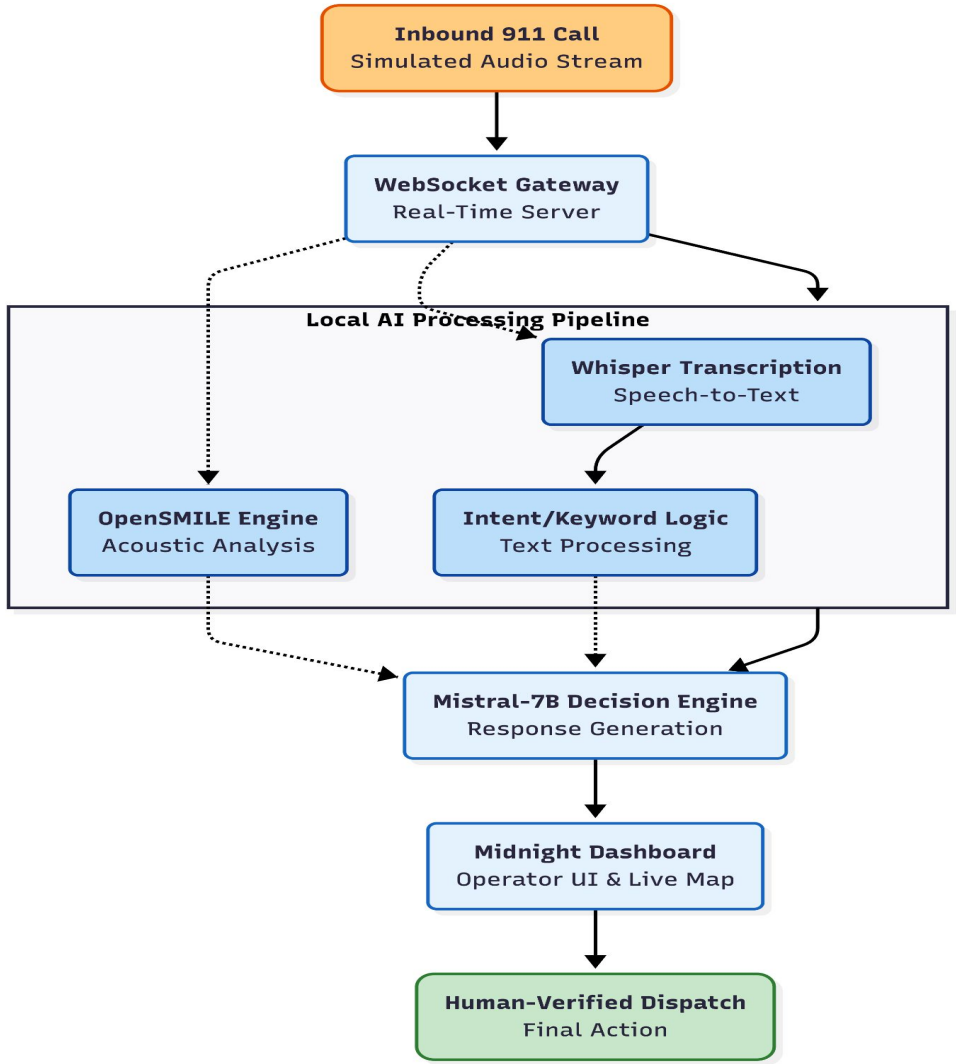
3- Project Scope

Scope Includes:

- Real-time call transcription (Whisper)
- Emotion detection using text & voice (BERT + OpenSMILE)
- Response generation using Fine-Tuned LLM
- Dispatcher dashboard with map and live chat
- Free open-source deployment (no paid APIs)

Scope Excludes:

- No real ambulance dispatch
- No Multilingual support
- No integration with real 911 systems (due to government control, cost, API limitations)



3- Project Methodology

- We adopted Agile methodology for flexibility and low risk.
- Project divided into modules: transcription, emotion detection, LLM, and dashboard.
- Weekly progress and testing handled in sprints across 30 weeks.
- Team Roles:
  - Abdullah Shaikh – Fine Tuning LLM & BERT
  - Murtaza Hassan – Backend(Fast API Server)
  - Muhammad Baqar – Frontend Dashboard & UI
  - Tahir Mehdi – Voice Pipeline & TTS Integration

4- Project Schedule

Phase	Timeline	Activity / Milestone	Resource (Owner)	Status
1. Requirements	Aug '25	SRS/SDS, Feasibility	Abdullah (Lead)	Completed
2. Architecture	Sep '25	Database (SQLite), API Design	Murtaza, Abdullah	Completed
3. Frontend	Oct '25	Midnight Dashboard, UI/UX	Abdullah, Baqar	Completed
4. Backend	Nov-Jan '25	FastAPI, WebSockets	Murtaza, Tahir	Completed
5. AI Integration	Feb '26	Whisper & OpenSMILE Setup	Tahir (All)	In Progress
6. Fine-Tuning	Mar-May '26	LLM / Training	Murtaza, Abdullah	Planned
7. Testing	May-June '26	UAT, Final Report	Baqar (QA), All	Planned

5- Project tools (or Hardware required)

Area	Tools Used
Programming	Python (Backend), TypeScript (Frontend)
AI Libraries	Faster-Whisper (STT), Fine-Tuned(LLM), OpenSMILE (Emotion)
Frontend	React, Next.js, Tailwind CSS
Backend/API	FastAPI, WebSockets (for real-time audio)
Dataset	911 calls (500) , SQLite (Lightweight, File-based)
TTS Engine	Mozilla TTS
Live Calling	WebSockets
Mapping	Leaflet.js (Offline-capable maps)
Hardware	Local NVIDIA GPU (Running generic CUDA)

6- Budget / Costing

Budget Breakdown

Item	Cost (PKR)
GPU Workstation (One-Time)	250,000
Peripheral Equipment	15,000
Software Licenses	0 (Open Source)
Total Estimated Cost	265,000

7- FYP Deliverables

- Operational Dispatcher Dashboard (Next.js & Real-time Maps).
- Local-First System Architecture (Design & Backend Logic) .
- Real-Time Audio Pipeline (WebSocket integration) .
- Emergency Triage Logic (Standard Operating Procedures) .
- Dataset & Documentation (SRS, SDS, User Manual) .

8- Reference

- AudEERING GmbH. "OpenSMILE Documentation." Munich: AudEERING, 2020.
- Huang, Jeremy, et al. "Whisper and ASR Systems." arXiv preprint arXiv:2009.xxxxx (2020).
- Rajput, D., and H. Singh. "BERT for Emotion Classification." Procedia Computer Science 218 (2023): 15-24.
- Reddy, Sandeep, and S. Kompalli. "Emotion Detection using Multimodal Analysis." Journal of AI & Ethics 3, no. 1 (2022): 45-58.